

IDEs for better designing

In the recent times, IDE software with feature-rich GUI for the embedded processor families has been the focus of many leading chip vendors. These employ the latest generation of mature code-generation tools and provide seamless, intuitive C/C++ and Assembly language editing, code-gen and debug supports, thereby making processor selection and product design more engineer friendly.

Subramanyam says, "Historically what used to happen was, if you were to create a sound bar or an AVR, you would tend to have the audio engineer and the software engineer sit together and try to choose the product. But in this case the graphical user interface offered for these DSPs enables the audio engineer to change the characteristics of various builders or the crossovers."

Additionally, the advanced algorithms for video processing have evolved for noise reduction as well as image formatting and conversion. The image enhancement algorithms add details to low-resolution images and adjust colour and contrast giving crisp, clear images on your display.

These advanced algorithms and encodings are reducing the computations, thereby reducing the power consumptions as well. Babu says, "On the way, people are trying to reduce power consumptions using accelerators, which are built using higher silicon technologies, and reducing the amount of data payload through advanced encodings."

Energy consumption

Managing the energy consumption is a major challenge for the application design technology today. Subramanyam says, "Energy consumption is a very important aspect of the consumer electronics industry today. The reason being, when you are watching the TV and you put it on standby, or if you put your audio system on standby, you would like the standby power of these devices to be very low. Most of the DSPs that we design have a full operational mode and then a standby

What are programmable accelerators?

The industry has been focussing on the hardware accelerators and programmable solutions separately. Hardware accelerators are circuits for computing specific algorithms. They provide very high performance and low power consumption. Hardware accelerators are used for computing standardised video algorithms (for example, video decompression) and some image-processing algorithms. But their huge gate count (silicon area) and no programmability cannot be ignored. Whilst we have multi-core CPUs with or without SIMD or VLIW co-processor extensions, DSPs, GPUs etc are programmable solutions which are primarily used for audio processing, but not so much for video. Although they have good programmability, their performance (acceleration) is low and power consumption is high. Sometimes such programmable solutions are used for certain computationally-intensive image processing/computer vision algorithms as well, despite the above drawbacks, because of the necessity of programmability.

But we now have programmable accelerators which provide high performance and low power consumption, just like hardware accelerators along with the programmability features of the solutions mentioned above.

"Our invention universal multifunction accelerator (UMA) has high performance and low power consumption, akin to hardware accelerators but with programmability of programmable solutions," says Dr Venu Kandadai, co-founder, Manjeera Digital Systems.

mode. The standby mode helps in saving power."

"We have ICs ranging from a few nanoamperes to a few milliamperes—it differs from product to product. And the market is pretty sensitive to the products which have sleep modes, hibernation modes, etc," he adds.

Other modules

We all know how important are analogue-to-digital converters (ADCs) and digital-to-analogue converters (DACs) for any signal processing system. These become essential while processing analogue signals such as voice or speech, and are used for enhancing voice processing features such as noise suppression, acoustic echo cancellation and multichannel beamforming. They also offer improved performance in voice capture processing such as voice control and recognition. Other applications such as digital set-top box systems, digital video camcorders, smartphones and tablets operating with digital signals are also designed with reliable, low-power and high-performance ADCs.

We have single packages for sampling, A/D conversion and anti-alias filtering, generating 24-bit values for both left and right inputs in serial form at sample rates up to 200kHz per channel. Most of the packages employ fifth-order, multi-bit delta sigma mod-

ulator followed by digital filtering and decimation, which removes the need for an external anti-alias filter, thereby reducing the number of components required for the designs. Further, audio/video codecs which combine audio ADCs and DACs into single ICs provide maximum flexibility, features and performance in the multimedia designs.

The risk factor

Ever-growing and highly-demanding entertainment and multimedia industry poses some risks as well for your designs. The primary one being the reliability and your commitment to future evolutions of the design. Although multi-vendor architecture is a plus point for the designs but a roadmap for the next-generation architectures and compatibility with the future parts will ensure improved integrations and reduced costs for your designs.

Nate Srinath, founder and director, Inxe says, "Every designer should try to mitigate business risk by adopting a multi-vendor capable architecture. The selection process should consider every A/V processor vendor's commitment and roadmap, coupled with technical and reference design support, along with proper software tools to mitigate business risk during developmental stages of the product." ●

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